

Variance-based sensitivity analysis and Shapley values

Variance-based importance measures

Importance measures and uncertainty quantification

- ▶ Key idea : can we quantify the amount of input uncertainty that creates dispersion in the output?
- ▶ Classical concept of dispersion : variance
- ▶ Importance of an input : amount of uncertainty (i.e., in terms of variance) it brings to the system.

ANOVA-Hoeffding decomposition (1948)

Starting point : $\mathbf{X} = (X_1, \dots, X_d)$ independent random variables

Theorem [Hoeffding decomposition]

- ▶ Let $F \in L^2(\mathbb{P}_X)$. Then F may be uniquely decomposed as the following orthogonal expansion

$$F(X) = \sum_{A \subset \{1, \dots, d\}} F_A(X_A) \text{ a.s. with an explicit expression of } F_A$$

- ▶ Consequence : $\text{Var}(F(X)) = \sum_{A \neq \emptyset} V_A$ with

$$V_A = \text{Var}(F_A(X_A)) = \sum_{B \subset A} (-1)^{|A|-|B|} \text{Var}(\mathbb{E}[F(X)|X_B])$$

Definition of Sobol indices (Sobol 1993)

Definition (normalized indices)

One defines for a given F

- ▶ Sobol index associated to A :

$$S_A[F] := \frac{\text{Var}(F_A(X_A))}{\text{Var}(F(X))} = \frac{V_A}{V}$$

- ▶ First order Sobol index associated to the input X_j :

$$S_j[F] := \frac{\text{Var}(F_j(X_j))}{\text{Var}(F(X))} = \frac{\text{Var}(\mathbb{E}[F(X)|X_j])}{\text{Var}(F(X))}$$

- ▶ Closed Sobol index associated to A :

$$S_{A,CI}[F] := \frac{\text{Var}[\mathbb{E}(F(X)|X_A)]}{\text{Var}(F(X))}$$

Definition of Sobol indices

Definition

One defines

- ▶ Total Sobol index associated to A [Homma and Saltelli, 1996]

$$T_A := 1 - \frac{\text{Var}(F_{\bar{A}}(X_{\bar{A}}))}{\text{Var}(F(X))}$$

- ▶ One can prove that

$$T_A := 1 - \frac{\text{Var}(\mathbb{E}[F(X)|X_{\bar{A}}])}{\text{Var}(F(X))}$$

- ▶ In particular, the total effect for $A := \{j\}$ is denoted T_j .

Definition of Sobol indices

First properties

1. If the function F does not depend on the random variable X_j , then $S_A[F] = 0$ for any A such that $j \in A$.
2. If $T_A = 1$, then F only depends on the random variables, the indices of which are in A .
3. It is clear that

$$1 = \sum_A S_A \text{ and } 0 \leq S_A \leq 1$$

Definition of Sobol indices

Interpretation

- ▶ $S_j[F]$ quantifies the part of the variability that is due to the action of variable X_j alone on F . We speak of the **first order importance** of X_j
- ▶ $S_{j_1,j_2}[F]$ quantifies the part of the variability that is due to the interaction between the variables X_{j_1} and X_{j_2} when the first orders have been removed.

Estimation of Sobol indices

The pick and freeze strategy (Jansen 1999, Saltelli 2002, Janon 2014...)

Estimation using naive Monte Carlo too costly

- ▶ Starting point of this strategy : **the knowledge of all closed indices** allows to recover all the other indices
- ▶ Focus on the estimation of the closed indices defined as

$$S_{A,CI} = \frac{\mathbb{E}(F(X)|X_A)}{\text{Var}(F(X))}$$

- ▶ Approach : the **so-called pick and freeze** strategy

Pick and Freeze strategy

- ▶ Basic idea of Pick and Freeze is that

$$S_j = \text{Corr}(Y, Y'),$$

where Y, Y' share the same X_j but are independent otherwise.

- ▶ That “shared- X_j , independent-otherwise” pair is obtained by *explicitly constructing* it
- ▶ That costs extra model evaluations and we need two samples.
- ▶ Improvements by Gamboa et al. 2022

See `Lecture1.ipynb` for implementation details

Main limitations of Sobol' indices

1. Requires **independent inputs features**. The ANOVA-Hoeffding decomposition assumes mutual independence; under correlation, Pick-and-Freeze estimators silently compute the **wrong quantity**.
2. **Purely global**. S_i is averaged over the whole input distribution; it cannot explain *one specific* prediction.
3. **Tied to variance**. Only ever answers “how much output *variance* does this explain?” — cannot adapt to another loss (accuracy, cost, ...) without leaving the framework entirely.

Shapley values to the rescue !

Definition of Shapley effects

General definition

- ▶ Origin (Shapley, 1953) : cooperative game theory
- ▶ A k -player game with the set of players $\mathcal{K} := \{1, \dots, K\}$ is defined as a real-valued function that maps a subset of $\mathcal{K}$ to its corresponding cost c
- ▶ The quantity $c(J)$ represents the cost that arises when the players in the subset J participate in the game

Definition of Shapley effects

General definition

- ▶ The Shapley value $\nu_j[c]$ of player j with respect to the cost function $c(\cdot)$ is defined as

$$\nu_j[c] := \sum_{J \subset \mathcal{K} \setminus \{j\}} \frac{(k - |J| - 1)! \cdot |J|!}{k!} (c(J \cup \{j\}) - c(J))$$

- ▶ $Sh_j[c]$ is the incremental cost of including player j in set J averaged over all sets J .
- ▶ Note that the sum contains **an exponential number** of terms

Definition of Shapley effects

Shapley values and global SA

In the context of global sensitivity analysis,

- ▶ Set of players : set of inputs of the model
- ▶ Choice of the cost function c ?

Definition of Shapley effects

Shapley values and global SA

Natural requirements for the cost function

- ▶ For each J , $c(J)$ measures the variance of Y caused by the uncertainty of the inputs in J
- ▶ $c(\emptyset) = 0$ and $c(K) = \text{Var}[Y]$.
- ▶ Some possible choices (Owen 2014)
 - ▶ Choice 1

$$c_1(J) = \text{Var}(\mathbb{E}[Y|X_J])$$

- ▶ Choice 2

$$c_2(J) = \mathbb{E}[\text{Var}(Y|X_{-J})]$$

Definition of Shapley effects

Theorem (Song 2016)

The Shapley values defined using cost function c_1 and c_2 are equivalent.

Shapley effect of the j -th input

Shapley value Sh_j obtained by applying the cost function c_1 or c_2 to the general definition

$$Sh_j := \nu_j[c_1] = \nu_j[c_2]$$

Definition of Shapley effects

Main property

Even if there is dependence or structural interactions among the inputs, one has

$$\sum_j Sh_j = Var(Y)$$

It is not the case for Sobol indices!

Estimating the Shapley effect

Two main questions

Chen, Covert, Lundberg & Lee, 2023) organize the literature around **two independent axes**:

- ▶ how “absent” features are handled?
- ▶ the definition of Shapley values involves an exponential sum.
How is it made tractable?

Estimating the Shapley effect

Feature removal approaches

- ▶ **Baseline Shapley values** : absent features replaced with a fixed reference x^b . Exactly computable, but the baseline choice is arbitrary.
- ▶ **Marginal (interventional) Shapley values** : absent features replaced by draws from their marginal distribution. Easiest to estimate; the most common variant.
- ▶ **Conditional Shapley values**: absent features drawn from the conditional distribution given observed features. Faithful to correlated features, but hard to estimate (empirical matching, parametric/Gaussian assumptions, generative or surrogate models).

Estimating the Shapley effect

Tractable estimation strategies in the model-agnostic case

The approaches are based on several equivalent mathematical characterizations

- ▶ The semivalue (SV) approach : Weighted average of marginal contributions over subsets sampled from a weighting distribution (ApproSemivalue; L-Shapley / C-Shapley)
- ▶ The random-order (RO) approach : Average marginal contribution across random permutations (IME; ApproShapley / permutation sampling)
- ▶ Shapley value as solution to a weighted least-squares regression : KernelSHAP, FastSHAP

Estimating the Shapley effect

Tractable estimation strategies in the model-specific case

In the case we know the black-box model, model-specific methods exploit its structure

- ▶ Linear models (LinearSHAP, Correlated LinearSHAP). In this case, the Shapley value has a closed-form solution which can be used to avoid sampling
- ▶ Tree models (Interventional TreeSHAP, Path-dependent TreeSHAP). A decision tree is a sequence of binary splits, that can be exploited, via a clever recursive algorithm, to compute the exact Shapley value in polynomial time
- ▶ Deep models (DeepLIFT, DeepSHAP, DASP, ShapNet). Each of these reuses the network's own forward/backward computational graph

An example : TreeSHAP

Algorithm 1 Estimating $E[f(x) \mid x_S]$

```
procedure EXPVALUE( $x, S, tree = \{v, a, b, t, r, d\}$ )  
  procedure G( $j, w$ )  
    if  $v_j \neq \text{internal}$  then  
      return  $w \cdot v_j$   
    else  
      if  $d_j \in S$  then  
        return  $G(a_j, w)$  if  $x_{d_j} \leq t_j$  else  $G(b_j, w)$   
      else  
        return  $G(a_j, wr_{a_j}/r_j) + G(b_j, wr_{b_j}/r_j)$   
      end if  
    end if  
  end procedure  
  return  $G(1, 1)$   
end procedure
```

Details about estimation in the model-agnostic case

Semivalue characterization (Castro 2009)

Sum over **subsets** S directly, weighted by how likely that subset size is under the classical Shapley weighting:

$$Sh_j(v) = \sum_{S \subseteq D \setminus \{j\}} P(S) (v(S \cup \{j\}) - v(S)),$$

with $P(S) = \frac{|S|! (|D| - |S| - 1)!}{|D|!}$

Details about estimation in the model-agnostic case

Semivalue sampling (ApproSemivalue): sample subsets, one player at a time

- ▶ To estimate Sh_j : repeatedly draw random subsets $S \subseteq D \setminus \{j\}$ (with probability $P(S)$), compute $v(S \cup \{j\}) - v(S)$ each time, and average.
- ▶ This is done **independently for each player**. Estimating Sh_1 tells you nothing directly about Sh_2

Details about estimation in the model-agnostic case

Random-order characterization.

- ▶ Sum over **permutations** π (orderings of all players)
- ▶ For each one look at who came before player i :

$$Sh_j(v) = \frac{1}{|D|!} \sum_{\pi \in \Pi(D)} (v(Pre^j(\pi) \cup \{j\}) - v(Pre^j(\pi)))$$

where $Pre^j(\pi)$ is the set of players appearing before j in ordering π .

Details about estimation in the model-agnostic case

Random-order sampling (ApproShapley): sample permutations, all players at once

- ▶ Draw one random permutation π of *all* players.
- ▶ Walk along it: start with $S = \emptyset$, and as each player is added one at a time, record their marginal contribution at that point in the walk.
- ▶ **One permutation gives a marginal contribution estimate for every player simultaneously** — for each player, one simply reads off what got added when they entered the ordering.

Details about estimation in the model-agnostic case

Song, Nelson & Staum (2016): estimating Shapley effects

- ▶ Apply the **random-order** (permutation) approach directly to the **variance-based** cost function from earlier:

$$v(S) = c_1(S) = \text{Var}(\mathbb{E}[Y \mid X_S])$$

- ▶ Same permutation-walk machinery as ApproShapley : sample a permutation π , walk along it, record the marginal contribution of each player as it enters
- ▶ The only thing that changes versus generic random-order sampling: here it is a variance of a conditional expectation, not a single model prediction

Details about estimation in the model-agnostic case

Why Song's approach is expensive

- ▶ Estimating it needs its own **nested Monte Carlo** procedure at *every step* of the permutation walk:
 - ▶ **Outer loop** — sample values of X_S
 - ▶ **Inner loop** — for each fixed X_S , resample X_{-S} to estimate the inner conditional expectation $\mathbb{E}[Y \mid X_S]$
 - ▶ Take the variance of the inner estimates across the outer draws
- ▶ Cost compounds: (many sampled permutations) $\times$ (a full nested double loop **per step, per permutation**)

Estimating the Shapley effect

Limits of Song's approaches

- ▶ Algorithm with a high computational cost
- ▶ Alternatives : consider surrogate models
- ▶ Leads to **LIME** and **KernelSHAP**

LIME

LIME (2016)

- ▶ Local surrogate models : interpretable models that are used to explain **individual predictions** of black box machine learning models.
- ▶ Local interpretable model-agnostic explanations (LIME) : approach for fitting surrogate models.
- ▶ Surrogate models trained to approximate the predictions of the underlying black box model.

LIME

LIME (2016)

- ▶ Explanation model for a given instance $\mathbf{x}$?
- ▶ Starting point
 - ▶ black-box model $\hat{F}$ trained from observations $\mathcal{D} := \{(x_i, y_i), i = 1, \dots, n\}$
 - ▶ white box/surrogate model $\hat{F}_{surrogate}$
- ▶ Local surrogate models with interpretability constraint : solution of the following optimization problem

$$\hat{F}_{surrogate}(\mathbf{x}) = \operatorname{argmin}_{g \in G} L(\hat{F}, g, \pi_{\mathbf{x}}) + \Omega(g)$$

where L loss, Ω complexity measure

LIME

LIME (2016)

The recipe for training local surrogate models:

- ▶ Select your instance of interest for which you want to have an explanation of its black box prediction.
- ▶ Perturb your dataset and get the black box predictions for these new points.
- ▶ Weight the new samples according to their proximity to the instance of interest.
- ▶ Train a weighted, interpretable model on the dataset with the variations.
- ▶ Explain the prediction by interpreting the local model.

LIME

LIME (2016)

- ▶ Linear regression can be chosen as an interpretable surrogate model.
- ▶ Number of features K to select
- ▶ Several methods for **training models with low complexity**, for e.g. Lasso.

KernelSHAP

KernelSHAP (2021)

- ▶ KernelSHAP : Linear LIME + Shapley values
- ▶ SHAP approach consider a weighted linear regression model as the local surrogate model and an appropriate weighting kernel without any penalization
- ▶ The regression coefficients of the corresponding LIME surrogate model estimates the SHAP values.
- ▶ A blog about KernelSHAP
<https://iancovert.com/blog/kernelshap/>

KernelSHAP

KernelSHAP (2021)

- Set for a given instance $\mathbf{x}$ and a given model F

$$v(S) = \mathbb{E}[F(\mathbf{x}_S, \mathbf{X}_{-S})]$$

- Search for coefficients that minimize

$$\hat{\beta} = \operatorname{argmin}_{\beta} \sum_{S \subseteq D} W(S) (u(S) - v(S))^2,$$

- $u(S) = \beta_{\emptyset} + \sum_{j \in S} \beta_j$ and $W(S) = \frac{|D| - 1}{\binom{|D|}{|S|} |S| (|D| - |S|)}$
- The approximation of the Shapley values are $\hat{\beta}_1, \dots, \hat{\beta}_d$, $\hat{\beta}_{\emptyset}$ is the output of the model without any features

KernelSHAP

Why this is not just a heuristic

- ▶ This is a **fourth exact characterization** of the Shapley value — solved *exactly* over all $2^{|D|}$ coalitions, $\hat{\beta}_j = Sh_j(v)$, with no approximation error
- ▶ $W(S)$ is **not** an arbitrary kernel choice: it is derived from the same combinatorial weighting $P(S) = \frac{|S|! (|D| - |S| - 1)!}{|D|!}$ that appears in the semivalue formula — the “Shapley kernel” is that weighting wearing a different hat

KernelSHAP

KernelSHAP

In practice **to avoid exponential computational complexity**

- ▶ The WLS problem above is **exact**; what is approximated is *how it is solved*: instead of summing over all $2^{|D|}$ coalitions, sample a moderate number of subsets according to $W(S)$
- ▶ Solve the (reduced) WLS problem on this sample instead of the full enumeration
- ▶ Implemented in the SHAP library

FastSHAP

FastSHAP (2022)

- ▶ KernelSHAP solves a **WLS problem for every new instance $\mathbf{x}$**
- ▶ It is accurate, but slow when many predictions need explaining (e.g. a live dashboard, auditing a whole test set)
- ▶ FastSHAP's idea: **amortize** this cost by training a single **explainer network $\phi_\theta(\mathbf{x})$** once, offline
- ▶ At inference time: **one forward pass** of ϕ_θ replaces solving a WLS problem from scratch

FastSHAP

Training objective

- ▶ Same weighted least-squares loss as KernelSHAP, but β is replaced by the **output of a network** evaluated at $\mathbf{x}$:

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathbf{x}} \mathbb{E}_{S \sim W(S)} \left[\left(\phi_{\theta}(\mathbf{x})^{\top} z_S + v_{\mathbf{x}}(\emptyset) - v_{\mathbf{x}}(S) \right)^2 \right]$$

where $z_S \in \{0, 1\}^{|D|}$ is the membership indicator of S

- ▶ The efficiency constraint $\sum_i \phi_{\theta}(\mathbf{x})_i = v_{\mathbf{x}}(D) - v_{\mathbf{x}}(\emptyset)$ is enforced at inference (additive normalization), not solved coalition-by-coalition
- ▶ Training samples many $(\mathbf{x}, S)$ pairs **once**; the network then generalizes to new, unseen $\mathbf{x}$

FastSHAP

FastSHAP vs. KernelSHAP

- ▶ **Speed:** KernelSHAP — many model queries *per instance*; FastSHAP — one forward pass per instance, after a one-time training cost
- ▶ **Exactness:** KernelSHAP converges to the exact WLS solution as the sample size grows; FastSHAP is only as good as the trained network — no convergence guarantee to the true $\phi_i(v)$
- ▶ **A structural difference:** KernelSHAP cleanly separates “estimate $\hat{v}(S)$ ” from “solve for ϕ ”; FastSHAP **fuses** the two — the network is trained end-to-end directly against the WLS objective, so there is no separate, swappable $\hat{v}$ stage
- ▶ **Bottom line:** FastSHAP trades exactness/guarantees for amortized, real-time explanation — the right tool when the same model must explain *many* instances, not a strictly better version of KernelSHAP

Estimating the Shapley effect (shapiq 2025)

Extension to groups of variables

For groups of variables

$$\mathcal{L}(\nu, \Phi_k) := \sum_{T \subseteq N} \mu(t) \left(\nu(T) - \sum_{S \subseteq T, |S| \leq k} \Phi_k(S) \right)^2$$

with

$$\mu(t) := \begin{cases} \mu_{\infty} & \text{if } t \in \{0, n\} \\ \frac{1}{\binom{n-2}{t-1}} & \text{else} \end{cases}$$